By considering perturbation theory up to the first order for a small value of λ and using Eq. (2), the energy eigenvalues of the Hamiltonian in Eq. (1) become (see appendix A),

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega + \frac{3\lambda\hbar^2}{4m^2\omega^2}(2n^2 + 2n + 1), \tag{3}$$

where $n = 1/(e^{\beta\omega} - 1)$ is the thermal occupation number and by using this value in Eq. (3), we calculate the average energies $\langle E\rangle_i$ $(i = A, B, C, D)$ of the oscillator during the four stages of quantum quartic Otto cycle as ($k_B = \hbar = m = 1$) (see appendix A):

$$\langle E\rangle_A = \frac{\omega_c}{2}\coth\left(\frac{\beta_c\omega_c}{2}\right) + \frac{3\lambda}{8\omega_c^2}\left[\coth^2\left(\frac{\beta_c\omega_c}{2}\right) + 1\right], \tag{4}$$

$$\langle E\rangle_B = \frac{\omega_h}{2}\coth\left(\frac{\beta_c\omega_c}{2}\right) + \frac{3\lambda}{8\omega_h^2}\left[\coth^2\left(\frac{\beta_c\omega_c}{2}\right) + 1\right], \tag{5}$$

$$\langle E\rangle_C = \frac{\omega_h}{2}\coth\left(\frac{\beta_h\omega_h}{2}\right) + \frac{3\lambda}{8\omega_h^2}\left[\coth^2\left(\frac{\beta_h\omega_h}{2}\right) + 1\right], \tag{6}$$

$$\langle E\rangle_D = \frac{\omega_c}{2}\coth\left(\frac{\beta_h\omega_h}{2}\right) + \frac{3\lambda}{8\omega_c^2}\left[\coth^2\left(\frac{\beta_h\omega_h}{2}\right) + 1\right]. \tag{7}$$

The amount of heat exchanged with the hot reservoir during the isochoric heating stage can be expressed as

$$\begin{aligned}\langle Q\rangle_h &= \langle E\rangle_C - \langle E\rangle_B \\ &= \frac{\omega_h}{2}\left[\coth\left(\frac{\beta_h\omega_h}{2}\right) - \coth\left(\frac{\beta_c\omega_c}{2}\right)\right] + \frac{3\lambda}{8\omega_h^2}\left[\coth^2\left(\frac{\beta_h\omega_h}{2}\right) - \coth^2\left(\frac{\beta_c\omega_c}{2}\right)\right],\end{aligned} \tag{8}$$

The heat released to the cold reservoir in the isochoric cooling stage is,

$$\begin{aligned}\langle Q\rangle_c &= \langle E\rangle_A - \langle E\rangle_D \\ &= \frac{\omega_c}{2}\left[\coth\left(\frac{\beta_c\omega_c}{2}\right) - \coth\left(\frac{\beta_h\omega_h}{2}\right)\right] + \frac{3\lambda}{8\omega_c^2}\left[\coth^2\left(\frac{\beta_c\omega_c}{2}\right) - \coth^2\left(\frac{\beta_h\omega_h}{2}\right)\right],\end{aligned} \tag{9}$$

When $\lambda \to 0$, the anharmonic contribution vanishes, and the results match the harmonic scenario.

We follow a sign convention where we take input (output) heat to be positive (negative). Therefore, from the first law of thermodynamics, the net extracted work from the engine is given by $W_{ext} = Q_c + Q_h$. In a quartic oscillator, the energy levels are more widely and unevenly spaced compared to a harmonic oscillator [60, 61]. Therefore, some transitions have very small energy gaps, while others have large gaps. Due to this complex energy structure, the system can accidentally jump to other states instead of smoothly staying in one energy state, leading to unwanted quantum transitions. This will give rise to irreversible losses that do not occur in simpler harmonic systems. Some energy gets wasted in the wrong quantum state and cannot be converted back to useful work, which reduces the amount of extracted work. We can also say that due to the unevenly spaced energy levels, the increase in energy during adiabatic compression is greater, but recovers less during expansion, reducing the net work output.

Now, the efficiency of an engine is the ratio of extracted work and input heat and is given by

$$\eta = \frac{W_{ext}}{Q_h}. \tag{10}$$

By utilizing Eq. (10), we determine the efficiency of a quantum quartic Otto engine as

$$\eta_{anh} = 1 - \frac{\omega_h^2}{\omega_c^2} - \frac{4\omega_h^2(\omega_c^3 - \omega_h^3)}{4\omega_h^3\omega_c^2 + 3\lambda\omega_c^2\left[\coth\left(\frac{\beta_c\omega_c}{2}\right) + \coth\left(\frac{\beta_h\omega_h}{2}\right)\right]} \tag{11}$$

In Fig. 2, the efficiency (Eq. (11)) is plotted against β_c for the quantum Otto engine, where the working substance is a quartic oscillator. The blue line indicates the efficiency of the engine corresponding to the harmonic case, while the red, green, and purple curves correspond to the efficiency of the quantum quartic engine characterized by $\lambda = 0.1$, 0.3, and 0.5, respectively. Similar to the behavior observed for work output, Fig.2 demonstrates that

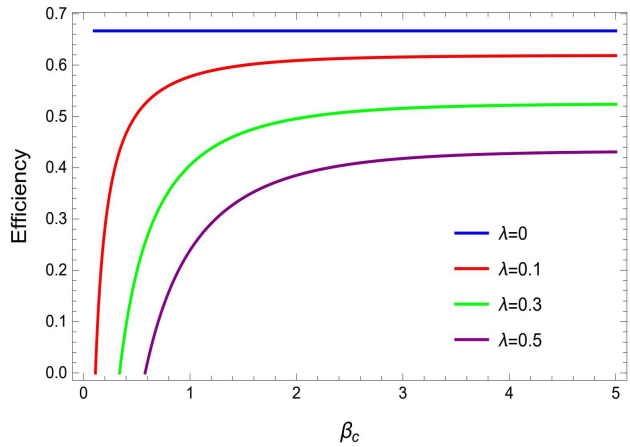

FIG. 2. Variation of efficiency with β_c for quantum Otto cycle with quartic oscillator as a working medium. The blue line represents the efficiency of the harmonic case ($\lambda = 0$), while the red, green, and purple curves imply the working medium as a quartic oscillator (QO) with $\lambda = \{0.1, 0.3, 0.5\}$. The other parameters are taken as $\beta_h = 0.5\beta_c$, $\omega_c = 2$ and $\omega_h = 3$.

the efficiency of the quantum quartic engine decreases as the anharmonicity parameter λ increases. The reason for the decrease in the efficiency of the engine is the complex energy level structure of the quartic oscillator, which causes unwanted quantum transitions, leading to degradation in the Otto cycle's work output and, therefore efficiency.

The efficiency, as defined in Eq. (11), can be approximated by its first-order Taylor series expansion as

$$\eta_{anh} = \eta_{har} + \lambda \cdot f(\omega_c, \omega_h, \beta_c, \beta_h) \tag{12}$$

where $\eta_{har} = 1 - \omega_c/\omega_h$ is the harmonic efficiency and $f(\omega_c, \omega_h, \beta_c, \beta_h) = -3(\omega_h^3 - \omega_c^3)\left[\coth\left(\frac{\beta_c\omega_c}{2}\right) + \coth\left(\frac{\beta_h\omega_h}{4}\right)\right] / (4\omega_c^2\omega_h^4)$.

Further, we study the variation in the coefficient of performance of a refrigerator with a quartic oscillator as a working material in the quantum Otto cycle. A refrigerator operates in reverse mode to that of an engine. For the refrigerator, $Q_c > 0$, $Q_h < 0$ and the invested work, $W_{in} = -(Q_c + Q_h) > 0$. The coefficient of performance for a refrigerator is given by

$$\zeta = \frac{Q_c}{W_{in}}. \tag{13}$$

With the help of Eqs. (8) and (9), COP is obtained as

$$\zeta_{anh} = \frac{\omega_h^2}{\omega_c^2 - \omega_h^2 + \dfrac{4\omega_c^2(\omega_h^3 - \omega_c^3)}{4\omega_c^3 + 3\lambda\left[\coth\left(\dfrac{\beta_c\omega_c}{2}\right) + \coth\left(\dfrac{\beta_h\omega_h}{2}\right)\right]}} \tag{14}$$

The first-order Taylor expansion of the efficiency given in Eq. (14) is expressed as

$$\zeta_{anh} = \zeta_{har} + \lambda \cdot g(\omega_c, \omega_h, \beta_c, \beta_h) \tag{15}$$

where $\zeta_{har} = \omega_c/(\omega_h - \omega_c)$ is the harmonic COP and $g(\omega_c, \omega_h, \beta_c, \beta_h) = 3(\omega_c^2 + \omega_c\omega_h + \omega_h^2)\left[\coth\left(\frac{\beta_c\omega_c}{2}\right) + \coth\left(\frac{\beta_h\omega_h}{4}\right)\right] / (4\omega_c^2\omega_h^2(\omega_h - \omega_c))$.

With the help of Eqs. (8), (9) and (13), we examine how the coefficient of performance of a refrigerator is affected when a quartic oscillator is used as a working substance in the quantum Otto cycle. Fig. 3 displays the analytically obtained COP as a function of β_c. The pink line corresponds to the harmonic case, whereas green, black, and orange curves represent the COP for quartic refrigerator with $\lambda = 0.1$, 0.3, and 0.5, respectively. It is evident that the COP increases as the anharmonicity parameter increases because of anharmonic distortion of the energy levels of quartic